

(a) textbook ABC-PMC, schematic:

one update per generation

shown: generation $t \rightarrow$
 $t + 1$, after 135 simulations

P_{t+1} : the N kept, weighted;
the rejected are discarded

(b) generation-free (ours):

one update per arrival

shown: the update after 84 evaluations

θ^* to a free worker; ρ^* arrives
later: append $(\theta^*, \rho^*, \epsilon_n, w^*)$

the current popula-
tion only; earlier gen-
erations are discarded

1
history

next threshold: a quan-
tile of the population's dis-
crepancies; accept or reject

2
bandwidth

every particle ever evaluated,
with its stored bandwidth and
weight

the k best form the archive;
a smooth kernel of width ϵ_n ,
which only ever tightens

weights carried over
from the last generation;
covariance from them

3
archive weights

one mixture proposal
over the whole population

4
proposal

propose, simulate, accept
or reject, one candidate at
a time, until N are kept

5
one draw

one mixture proposal over the
archive, rebuilt at every call

one candidate: a parent drawn
by weight, then perturbed

the kept are weighted
against that one pro-
posal, after acceptance

6
weight

weighted against the current
proposal and the S stored pro-
posals, before it is simulated

barrier: q_{t+1} needs the complete P_{t+1}

fewer than k particles: prior
draw, $w^* = 1$, no bandwidth
stored, straight to a worker